

IMPLICATIONS OF SPATIAL HETEROGENEITY FOR CATASTROPHIC REGIME SHIFTS IN ECOSYSTEMS

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Abstract. Although alternative stable states are commonly found in simple models, it seems reasonable to assume that the response of real ecosystems to environmental change should often be smoothed by spatial heterogeneity and other stabilizing mechanisms. Here, we systematically explore the effect of spatial heterogeneity on regime shifts for three different models, which we run on a one-dimensional lattice with different spatial distributions of an environmental factor (e.g., soil fertility, water level). If dispersion between patches is negligible, the response to gradual change in some overall stressor (e.g., precipitation, nutrient load) is straightforward. Because of the environmental heterogeneity, each patch shifts to the other stable state at different values of the overall control variable. Therefore, the response of the ecosystem as a whole is gradual (i.e., the average of many asynchronous small shifts) instead of catastrophic. However, in response to a reverse change in the global stressor, the system always shows hysteresis, as each individual patch shifts back to the original state at a different value of the control parameter than the critical threshold for the forward shift. If dispersion between patches occurs, the response to change in the overall control parameter becomes dependent on the spatial pattern of environmental heterogeneity. If the environmental parameter is randomly distributed in space, the overall response tends to remain surprisingly catastrophic, and hysteresis is hardly reduced as compared to the homogeneous case. By contrast, in a smooth environmental gradient, the response of the overall system is gradual, and hysteresis is much smaller. In fact, hysteresis is largely reduced to the initial phases, in which none of the patches have shifted to the alternative state yet. As soon as the first patch shifts, a domino effect occurs, pushing over the neighboring patches. In conclusion, our results suggest that spatial heterogeneity may weaken the tendency for large-scale catastrophic regime shifts if dispersion is unimportant or if local environmental characteristics vary along a smooth gradient.

Key words: *hysteresis; lattice model; mobile link organisms; multiple attractors; regime shifts; spatial heterogeneity.*

INTRODUCTION

Omnipresent global change implies that virtually all ecosystems are exposed to gradual, often almost linear change in external conditions such as climate, inputs of nutrients or toxic chemicals, groundwater reduction, habitat fragmentation, or harvest (Vitousek et al. 1997, Tilman and Lehman 2001). Although this is generally assumed to result in gradual change in the ecosystems, occasionally sudden drastic regime shifts may occur (e.g., Steele 1998, Hare and Mantua 2000). One possible explanation for such shifts is that the ecosystem has alternative stable states (Carpenter 2001, Scheffer et al. 2001). This implies that under the same external conditions the system can be in two or more stable states. Obviously, “stable state” is a simplification, as internally generated dynamics and environmental stochasticity will always cause fluctuations (Scheffer and Carpenter 2003). Nonetheless, we will use the term

here for simplicity. The phenomenon of alternative stable states is important for two reasons (Scheffer et al. 2001). Firstly, it implies that subjected to a slowly changing external factor (such as climate), an ecosystem may show little change until at a critical point, a sudden shift to an alternative contrasting state occurs. Such a shift is often triggered by a stochastic event, but loss of resilience (*sensu* Holling 1973) is the underlying reason. Secondly, the presence of alternative stable states implies that if a system has gone through such a state shift, it tends to remain in the new state until the control variable is changed back to a much lower level, a pattern known as hysteresis.

Up till now, theory on these phenomena has been largely based on simple models (e.g., Volterra 1926, Noy-Meir 1975, May 1977, Tilman 1977, Scheffer 1998) in which the state of the ecosystem is described by a few differential equations at most. These models are easy to analyze, but leave out much of the potentially important complexity of a real ecosystem. In fact, one could wonder whether the occurrence of an alternative stable state could be an artifact of oversimpli-

Manuscript received 31 March 2004; revised 28 September 2004; accepted 17 November 2004. Corresponding Editor: W. S. C. Gurney.

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TABLE 1. Models and parameters used in this study.

Model and parameters	Definition or value	Description
$\frac{dX_i}{dt} = X_i \left(1 - \frac{X_i}{K_i} \right) - c \frac{X_i^p}{X_i^p + 1} + d(X_{i-1} - 2X_i + X_{i+1})$		grazing of a logistically growing food (X).
K_i	carrying capacity, the spatial heterogeneous parameter at patch i (range: 5–10)	
c	maximum grazing rate, the slow control parameter (1–3)	
p	2	
$\frac{dX_i}{dt} = a_i - X_i + r \frac{X_i^p}{X_i^p + 1} + d(X_{i-1} - 2X_i + X_{i+1})$		nondimensional version of a model of nutrient (X) cycling in lakes
a_i	nutrient load, the spatial heterogeneous parameter at patch i (range: 0.25–0.5)	
r	maximum recycling rate, the slow control parameter (range: 0–8)	
p	exponent of the Hill function (4)	
$\frac{dE_i}{dt} = r_E E_i \left(1 - \frac{E_i h_v + V_i}{E_o h_v} \right) + d_E(E_{i-1} + 2E_i + E_{i+1})$		macrophytes (V) and vertical light attenuation (E) in lakes
$\frac{dV_i}{dt} = r_V V_i \left(1 - V_i \frac{h_{E,i}^p + E_i^p}{h_E^p} \right) + d_V(V_{i-1} + 2V_i + V_{i+1})$		
$h_{E,i}$	spatial heterogeneous parameter (1–3)	
E_o	slow control parameter (1–12)	
r_E	0.1	
r_V	0.05	
h_v	0.2	
p	4	

fication. One of the obvious oversimplifications in most models so far is the neglect of the fact that real ecosystems usually are spatially heterogeneous (Scheffer and Carpenter 2003). Many ecosystems consist of patches of habitats and certain keystone organisms may actively move between these areas (“mobile link organisms,” see Gilbert [1980], Lundberg and Moberg [2003]). Though several authors stress the potential importance of spatial heterogeneity and exchange in the environment for alternative stable states (Connell and Sousa 1983, Petraitis and Latham 1999, Levin 2000, Peterson 2000, Nystrom and Folke 2001) few studies actually explore the topic experimentally or with models.

Here, we use simple models to analyze how spatial heterogeneity in the environment and movement of matter and organisms should be expected to affect the likelihood of large-scale catastrophic transitions and hysteresis in ecosystems.

MODELS

For our analyses, we used three different models that are known to have alternative stable states.

The first model describes grazing of a fixed number of grazers on a logistically growing species (Noy-Meir 1975, May 1977). It has been applied to describe overexploitation vs. underexploitation of vegetation or prey for various different ecosystems. Examples are the dy-

namics of spruce budworms (May 1977, Ludwig et al. 1978), exploitation of fish communities by humans (May 1977, Steele and Henderson 1984), overgrazing in arid systems (Van de Koppel et al. 1997), and harvesting of macrophytes (Van Nes et al. 2002).

The second model is an equation by Carpenter et al. (1999) describing phosphorus cycling in stratified lakes. Specifically, it focuses on the loss of phosphorus from the epilimnion and the sudden recycling if the hypolimnion becomes anoxic. A difference with the other models is that the state variable is abiotic. The same equation can also be used to describe colonization of barren soils in drylands by vegetation (Scheffer et al. 2000). In this case, the state variable is amount of barren soil.

The third model is a somewhat more specific model of two coupled differential equations by Scheffer (1998), describing the relation between macrophytes and turbidity in shallow lakes. The basis of this model is the well-studied positive feedback of aquatic vegetation on their own growth. Vegetation reduces turbidity, and low turbidity enhances vegetation.

These three models were chosen, as they are simple and generic representatives of the models that were used to develop the alternative states theory in various ecosystems. All equations and parameter settings are given in Table 1. To analyze the consequences of spatial heterogeneity, we simulated the behavior of these mod-

els in a one-dimensional lattice. We studied the effects of two aspects: the spatial pattern of heterogeneity and the occurrence of dispersion of the studied populations in space.

Dispersion of the state variable in continuous space may be modeled by adding a dispersion term to each differential equation. In case of a one-dimensional space (z), one obtains partial differential equation (see, e.g., Okubo 1980) of the form

$$\frac{\partial X}{\partial t} = f(X_i) + D \frac{\partial^2 X}{\partial z^2}$$

in which $f(X_i)$ is the right-hand side of the differential equations of each model and D is the dispersion coefficient. In our case with a series of n linearly coupled, equally spaced patches, the formulation becomes the following (Levin 1976, Keitt et al. 2001):

$$\frac{dX_i}{dt} = f(X_i) + d(X_{i+1} - 2X_i + X_{i-1}) \quad 2 < i < n - 1$$

in which $f(X_i)$ is the right-hand side of the differential equations of each model (see Table 1). The dispersion coefficient D is replaced by the discrete analog d . At both ends of the array of patches, we assumed that there was exchange with one neighboring cell only. In the model with two coupled differential equations, the exchange between patches of both state variables was treated similarly. Both dispersal of matter and organisms were treated as diffusion processes. Of course, for dispersal of organisms this is a simplification, but many studies have shown that the collective behavior of organisms can be approximated as a diffusion process (Levin 1992). We did not study the effects of other types of dispersal, even though pattern formation in spatial models can be sensitive to the choice of dispersal kernels (Anderson and Neuhauser 2002).

Since we aimed to study the effect of a gradually increasing stress to the system in a spatially heterogeneous setting, one parameter of each model was arbitrarily chosen to express spatial heterogeneity of the landscape (e.g., topographic situation, or soil richness). This parameter is set differently for each patch. A second parameter (the “control parameter”) was chosen to represent a slowly changing stressor (e.g., nutrient loading to a lake, or rainfall in a terrestrial setting), which acted equally for all patches. For each model, we studied the effect of three spatial settings: (1) no spatial heterogeneity, (2) a linear gradient in the spatial parameter along the lattice, and (3) a randomly drawn value of the spatial parameter for each patch to produce uncorrelated environmental variation. In each of these settings, we studied the effect of three levels of dispersion: (1) none, (2) moderate, and (3) strong. The analyses of the randomly drawn spatial patterns were repeated 100 times to obtain an impression of the effect of stochasticity. We show the 0.05, 0.5, and 0.95 percentiles of these 100 runs.

Additionally, we checked briefly whether the generated patterns were sensitive to the assumption of one-dimensional dispersion. We implemented the model of Scheffer (1998) also on a two dimensional grid of 50×50 cells. Each cell was now connected to its four neighbors in a similar way as described above. For comparison with the one-dimensional model, we divided the two-dimensional dispersion coefficient by two, as dispersion is now in four directions instead of one.

We implemented the models using the software package GRIND for MATLAB (*available online*),² which uses a Runge-Kutta method to solve the differential equations. For a bifurcation analysis of the original models, we used the bifurcation continuation program LOCBIF (Khibnik et al. 1992). In modeling studies, regime shifts are usually called “catastrophic bifurcations” and stable states “attractors,” but in this paper, we will use the ecological naming.

ANALYSES AND RESULTS

Due to limitations of the bifurcation software, we used a numerical procedure to scan the tendency for catastrophic transitions and hysteresis in response to a gradual increase and a subsequent decrease in a hypothetical stressor (the control parameter) of each model. The control parameter was increased in 50 small steps. After each of these increments, the models were simulated in 1000 time steps, which was enough to allow stabilization. The mean biomass (i.e., the value of a state variable) over the spatial gradient was then plotted. After the 50 incremental steps, the last achieved value of the state variable was kept as the new starting condition and the control parameter was changed back in 50 steps, following the same procedure on each step.

Although there are pronounced differences in the way in which the three models responded to the global control parameter in different spatial settings (Figs. 1–3), some general patterns emerge.

Without dispersion (Figs. 1–3b, c), the spatial correlation of heterogeneity (smooth gradient or random) is irrelevant. Each cell is independently responding to the control variable. Due to the environmental heterogeneity, each cell shifts to the alternative stable state at a different value of the control variable. As a result, the averaged response of the system is gradual (or more precisely, consists of many small steps). Obviously, there always remains a pronounced hysteresis in the response of the system as each isolated individual cell has the hysteresis characteristic of the non-spatial system (Figs. 1–3a).

With dispersion, the pattern depends strongly on the spatial correlation of environmental parameter. If environmental conditions vary randomly in space (Figs. 1–3e, g) we are back to pronounced hysteresis, and

² <http://www.dow.wau.nl/aew/grind/>

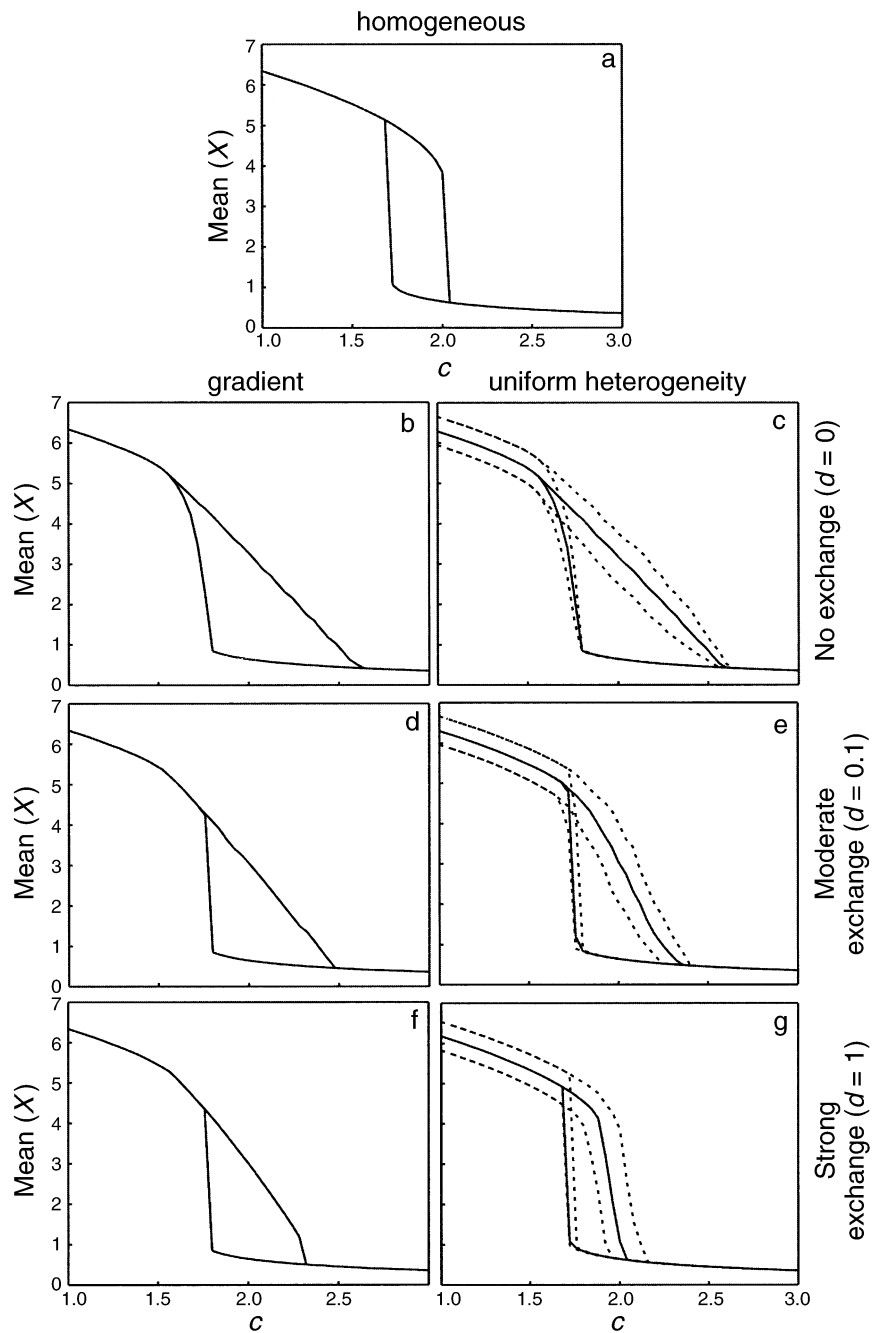


FIG. 1. The effect of spatial heterogeneity (K , carrying capacity, is between 5 and 10) on hysteresis of the model of May (1977); c is the maximum grazing rate. (a) Hysteresis in the original model ($K = 7.5$); (b) gradient without dispersion; (c) random heterogeneity without dispersion (0.05 [dotted line], 0.5 [solid line], and 0.95 [dotted line] percentiles of 100 runs); (d) gradient with moderate dispersion; (e) random heterogeneity with moderate dispersion (0.05 [dotted line], 0.5 [solid line], and 0.95 [dotted line] percentiles of 100 runs); (f) gradient with strong dispersion ($d = 1$); (g) random heterogeneity with strong dispersion (0.05 [dotted line], 0.5 [solid line], and 0.95 [dotted line] percentiles of 100 runs).

sharp synchronized catastrophic shifts of the entire system. By contrast, if there is a smooth environmental gradient in space (Figs. 1–3d, f), the response of the system tends to be gradual, and hysteresis becomes reduced if dispersion is strong. In this case, a domino effect can occur. The first cell that can switch to the

alternate stable state is delayed due to the continuous exchange with neighboring cells, but as soon as one cell switches to the other equilibrium, the neighboring cells then shift quickly too (Fig. 3f is a clear example).

Some eye-catching deviations from the general patterns occur. For instance, in the model of May (1977),

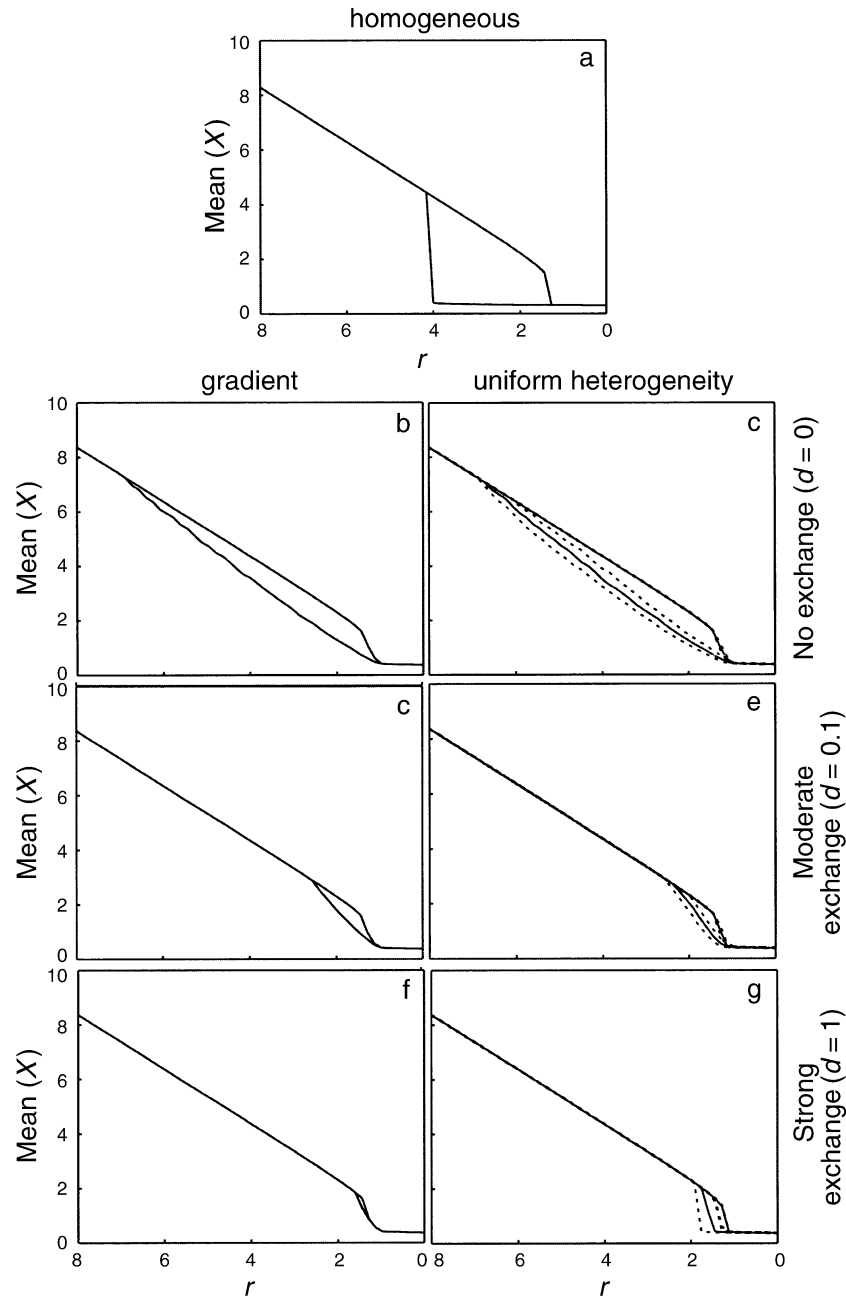


FIG. 2. The effect of spatial heterogeneity (a , nutrient load, is between 0.25 and 0.5) on hysteresis of the Carpenter et al. (1999) model; r is the maximum recycling rate. The description of panels is the same as in Fig. 1.

there seems to be hardly any effect of spatial heterogeneity and dispersion on the recovery of the state variable X (lower state shift). It occurs always in a catastrophic way at a c of about 1.75. Similarly, the upper branch of the Carpenter et al. (1999) model has always the same shape and the state shift always occurs at an r of approximately 1.3. Though this may seem puzzling at first sight, there is a simple explanation, which can be visualized by plotting two-dimensional bifurcation plots of the original models without spatial

heterogeneity (Fig. 4, right-hand panels). The parameter values where a state shift takes place are shown in these figures. The global control parameter is on the x -axis and the parameter for which we are studying spatial heterogeneity is on the y -axis. In both cases, the parameter that determines spatial heterogeneity has a much stronger effect on one of the two bifurcations. In the May (1977) model, the carrying capacity (K) has little effect on the shift to the high- X equilibrium. In the Carpenter et al. (1999) model, the shift to the

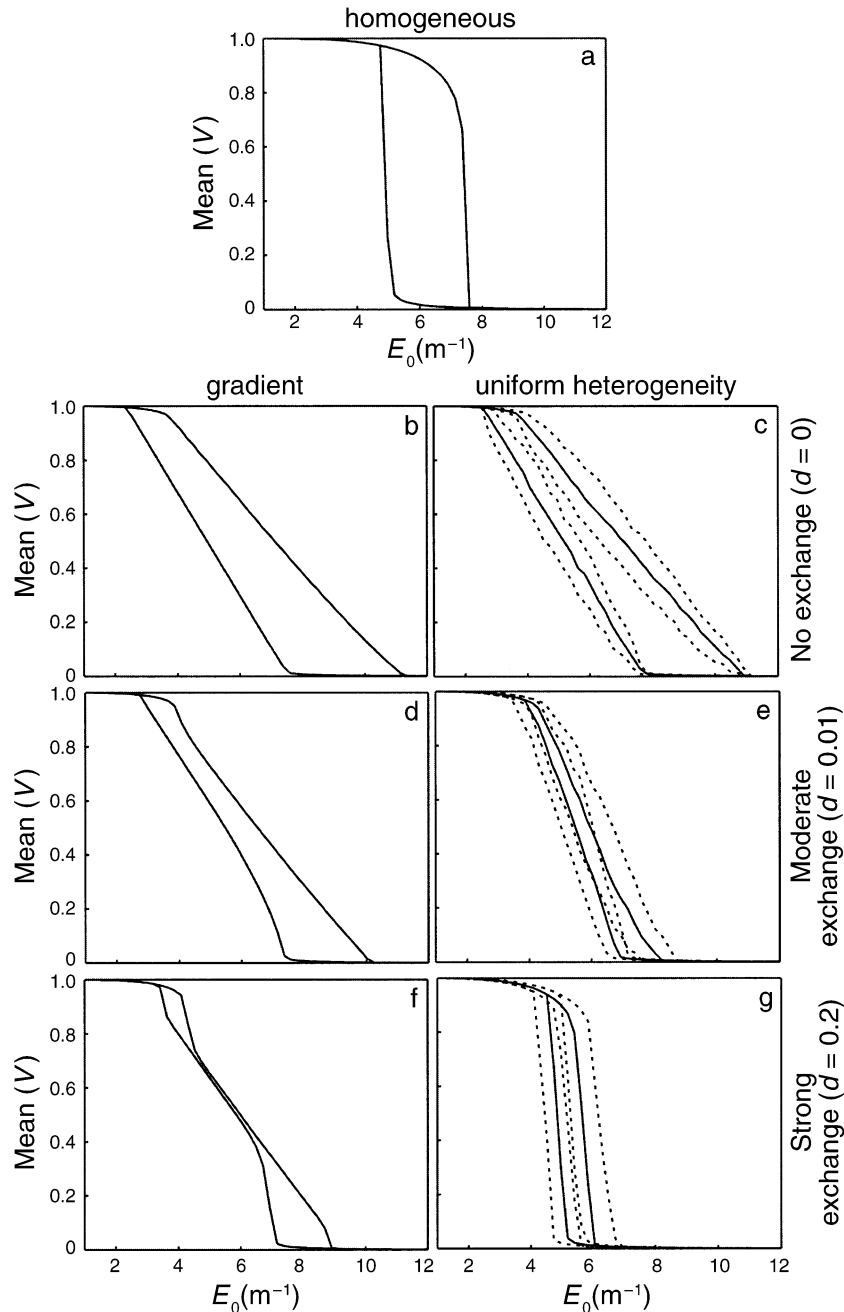


FIG. 3. The effect of spatial heterogeneity (h_E between 1 and 3) on hysteresis of the vegetation model of Scheffer (1998). E_0 is the vertical light attenuation of the lake if there is no vegetation. The description of panels is the same as in Fig. 1.

low- X equilibrium occurs always at the same r , independently of the spatial heterogeneous parameter a . Thus, in both cases the switch to the other equilibrium is synchronized, as the spatial heterogeneity has little effect on one of the two bifurcation points. Phrased in biological terms, spatial heterogeneity in a factor that is, for instance, important for collapse but irrelevant for recovery, makes the overall pattern of recovery gradual, but leaves that of collapse abrupt.

It is also remarkable that hysteresis in the Carpenter et al. (1999) model is reduced much more strongly than in the other models. This can be understood to be a result of the shape of the catastrophe fold, which is more asymmetric than the other models (Fig. 4, left-hand panels). The low-phosphorus stable state is always close to the unstable dotted equilibrium that marks the border of the basin of attraction. Thus, this state has a lower resilience than the high-phosphorus

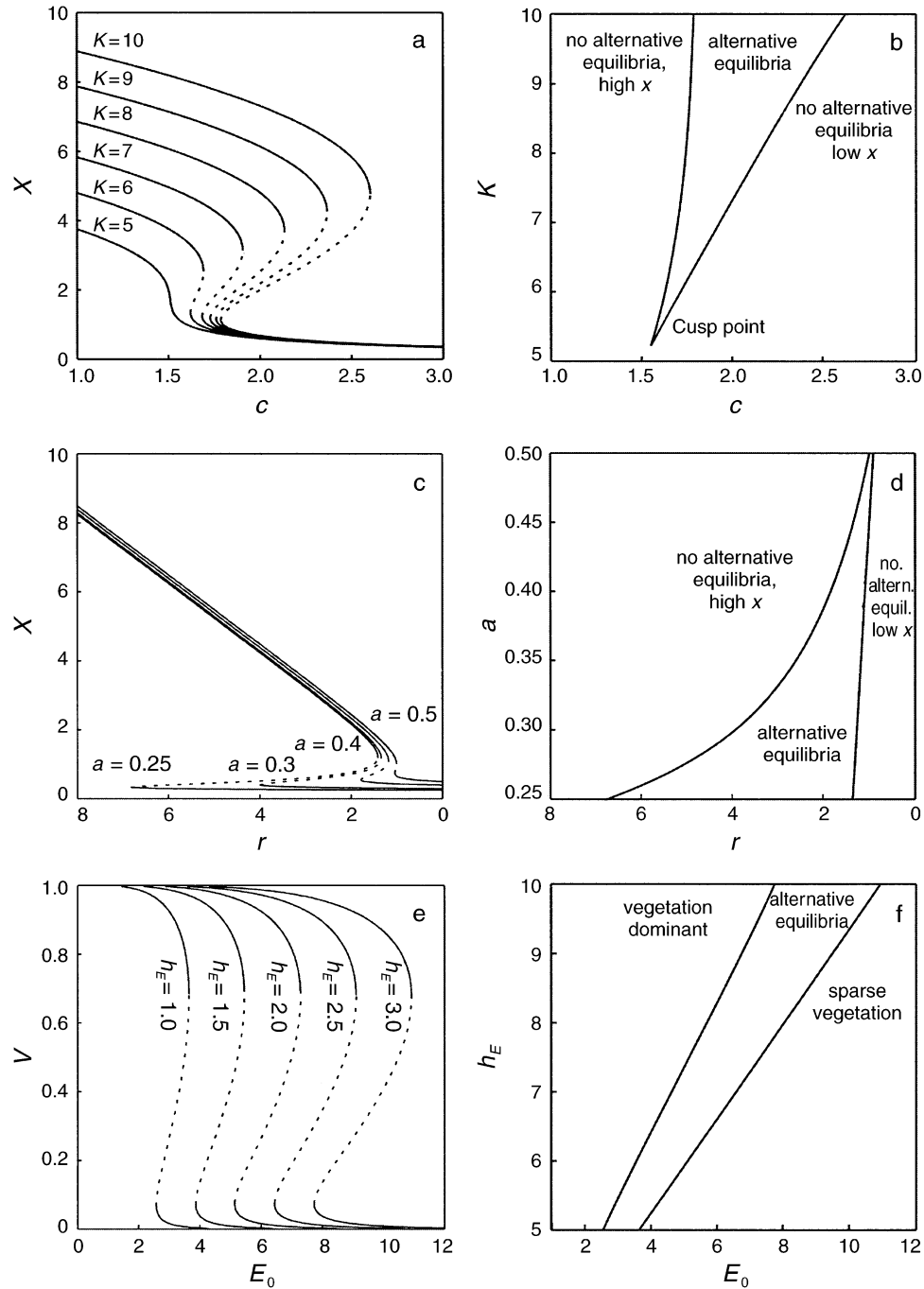


FIG. 4. Bifurcation plots of the models without spatial heterogeneity. Left-hand panels show catastrophe folds, and right-hand panels show two-dimensional bifurcation plots: (a and b) the May (1977) model; (c and d) the model of Carpenter et al. (1999); (e and f) the model of Scheffer (1998). The cusp point is the point where two fold bifurcations meet. Beyond this point, there are no alternative stable states.

state, whereas in the other models both states have a similar resilience. Therefore, even a tiny inflow from a neighboring patch can make a patch in the lower phosphorus equilibrium switch to the other equilibrium. For instance, simulation experiments showed that the upper equilibrium could handle at least a 10-fold

higher import of X before it switches to the other equilibrium (not shown). Phrased in general terms, if one of the alternative states is always very fragile, spatial heterogeneity favors the more stable alternative state.

The analysis seems not to be sensitive to the fact that we used a one-dimensional grid. We implemented

the Scheffer (1998) model in a two dimensional (50×50) grid and did a numerical bifurcation analysis similar as the one-dimensional case (Fig. 3e). The results were highly similar (Fig. 5a). A thorough analysis of the self-organized pattern formation in this model is beyond the scope of this paper. Nonetheless, as an illustration, we show a generated pattern in this model. Starting with homogeneous vegetation in a heterogeneous environment (Fig. 5b), we simulated 2000 days with moderate dispersion and generated a stable patchy pattern.

DISCUSSION

Our analyses show that the particular spatial organization of ecosystems may have pronounced implications for the likelihood that global change in environmental factors leads to a large-scale catastrophic state shift. The effects we see on multiple stable model systems placed in an artificial landscape allow three specific predictions:

1) Heterogeneity in combination with little dispersion will cause the response of a system to be localized. Averaged over the landscape, this implies gradual rather than sudden ecological change. Large-scale regime shifts will be restricted to systems that are relatively homogeneous and/or characterized by relatively intense dispersion of matter or organisms.

2) As forward and backward state shifts may depend on different mechanisms, one can be localized while the other occurs synchronized over a large scale.

3) Hysteresis is preserved in all cases except when spatial heterogeneity is aligned along a smooth gradient.

We are not aware of any empirical work related to the latter prediction. However, a number of field studies may be interpreted in the light of the first two predictions. We first discuss this work, and then link our work to recent theories on the effect of mobile link species and of self-organized patchiness on ecosystem resilience.

The idea (prediction 1) that dispersal can make heterogeneous systems behave like single homogeneously mixed systems is intuitively straightforward, and has been shown in other contexts in earlier theoretical analyses (Levin 1974, Scheffer and De Boer 1995, Loreau et al. 2003). The interesting question is whether we can find out which dispersal rates would be critical to make a system behave in synchrony over larger scales. Obviously, the impact of dispersal should depend on the dispersal rates relative to the rates of local processes. However, in natural systems we typically deal with many interacting variables (species, substances etc.) with different characteristic rates of dispersal and local dynamics. Therefore, it is not obvious how the predictions derived from simple models should work out quantitatively for real ecosystems. There are a few examples that may illustrate the range of dynamics that can be found in nature.

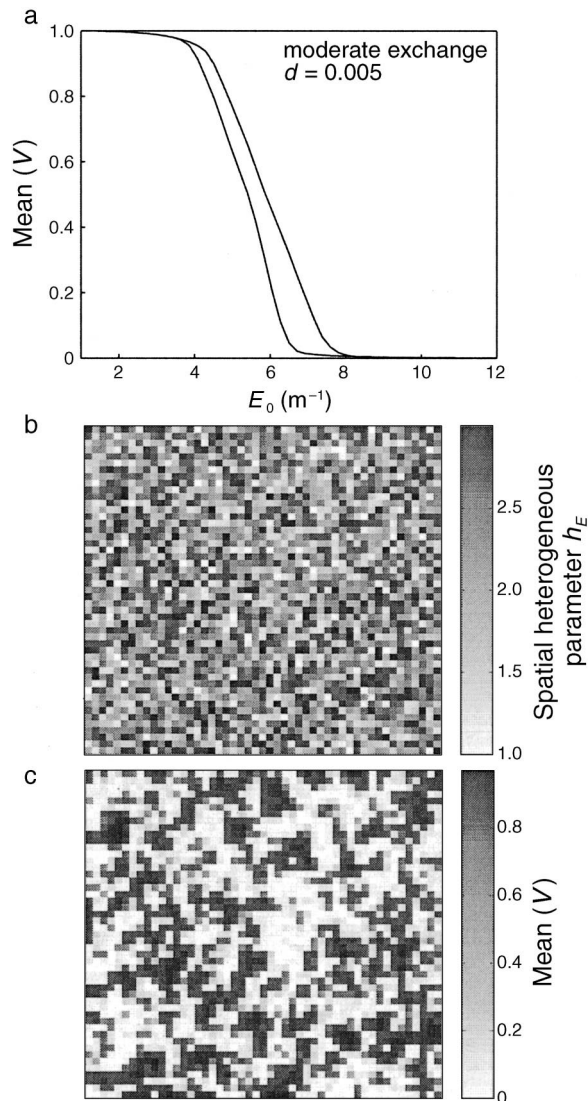


FIG. 5. Two-dimensional (50×50 grid) implementation of the Scheffer (1998) model. (a) The effect of random spatial heterogeneity on the hysteresis of the model; (b) the variation of the spatial heterogeneous variable; (c) the resulting pattern after stabilizing at $E_0 = 6$ (initial conditions: both E and V are 0.01 in all cells).

Large-scale synchronized shifts

Oceans and lakes are perhaps the most obvious examples of systems with large dispersion. Indeed, synchronous shifts over large areas have been described for oceans. A dramatic example from the remote past is the occurrence of massive marine anoxic events (Algeo and Scheckler 1998). These episodes of large-scale phosphorus release from oceanic sediments may represent an alternative state in which anoxic conditions are promoted by high algal production which in turn benefits from the phosphorus release (Van Cappellen and Ingall 1994). In more recent history, communities of the Northern Pacific are known to shift occasionally

in synchrony from one rather stable condition to another one over very large scales (Hare and Mantua 2000). Perhaps the best studied example of ecosystems shifting as a whole between alternative states are shallow lakes (Scheffer 1998). Here, eutrophication is a major agent invoking an often-sudden shift from a clear water state dominated by submerged vegetation, to a turbid state dominated by phytoplankton.

Locally decoupled alternative states

Remarkably, despite apparent exchange of matter and organisms in lakes, there are also examples where co-occurrence of contrasting states is observed. For instance, Cottenie et al. (2001) found that different alternative states can persist in neighboring ponds of a system of 33 similar interconnected ponds, despite free exchange of water and possibility for organisms to migrate.

Perhaps the most striking example of co-occurrence of alternative states in a seemingly mixed system is the eutrophied shallow Lake Veluwe. For many years, the clarity of the large turbid lake hardly responded to a reduction in nutrient loading, until surprisingly local areas with clear water and stonewort vegetation started developing (Van den Berg et al. 1998). The areas with clear water gradually expanded over the years but left a large turbid water region in the rest of the lake, until an apparently critical situation was reached that triggered a large-scale shift to a clear water state of the entire lake (Meijer 2000).

Scale difference in forward and backward shifts

Our analyses illustrate that forward and backward shifts may be differentially sensitive to mechanisms and parameters, and that this may have implications for the scale of the two shifts. For instance, in the May (1977) model, carrying capacity of local patches has little effect on the switch from underexploitation to overexploitation. Therefore, this shift always takes place synchronized whereas the backward shift occurs on local scales. There are indeed examples of such scale differences in forward and backward shifts. The most striking ones to our knowledge deal with large-scale pest outbreaks. For instance, a rinder pest epidemic decimated most large herbivores in much of the South of Africa leading to large scale regeneration of woodlands for a few decades from the 1890s onward (Walker 1989, Dublin et al. 1990). The grazers only affect the forward shift as they can control seedling development, but once established the woodlands cannot be eliminated by grazers. The backward shift (woodland destruction) happens on local scales by humans and sometimes-high densities of elephants. It has been proceeding gradually, and is probably irreversible unless herbivore numbers (controlling the forward shift) are again reduced (Walker 1989). Another example of a large scale shift triggered by a pathogen is the collapse of coral reef communities in the Caribbean (Knowlton

1992, McCook 1999, Nystrom et al. 2000, Bellwood et al. 2004). In 1982–1984, a species specific pathogen caused mass mortality of the sea urchin *Diadema antillarum* with far reaching consequences, as illustrated by the changes documented on Jamaican reefs (Hughes 1994). Urchin densities crashed to 1% of the original level, and this triggered a complete change in the reef community. Rapidly the reefs became overgrown by brown fleshy algae that were now released from grazing by urchins and the entire community of the reefs changed profoundly. So far, the backward shift to the coral dominated state is only observed locally.

Mobile links and resilience

Organisms that actively connect habitats are sometimes called “mobile links” (Gilbert 1980). In a recent review, Lundberg and Moberg (2003) argue that mobile links contribute to ecosystem resilience as they provide ecological memory (sensu Peterson 2002). If a site is disturbed, transport from an undisturbed support area could make the site switch back to the original state. Our results suggest that this ‘repair effect’ of mobile species on resilience may not always work out that way. In fact, if the system is already vulnerable to collapse dispersion (mobility) may actually invoke a domino effect causing a rapid spread of the disturbance rather than a local repair. The pathogen spread mentioned earlier is in a sense an example of such domino effects of mobile links.

Self organized heterogeneity and regime shifts

It is important to stress that we have dealt with a particular subclass of spatial ecological models. We looked at models that have alternative attractors, subjected to random spatial dispersion. So-far most work has been on models that have no alternative attractors in a non-spatial setting, but have non-random exchange of matter allowing for self-organized pattern formation (e.g., McCauley et al. 1993, Peterson 2002, Rietkerk et al. 2004). Perhaps the most well-known example in ecosystems is the formation of self-enhancing vegetated “islands of fertility” (Schlesinger et al. 1990) in arid regions. These patches benefit from neighboring unvegetated areas from which they receive rainwater and nutrients transported by surface run-off that then infiltrates in the vegetated islands. The resulting local positive feedback in the development of the islands combined with the negative feedback due to competition with other patches on a larger scale produces characteristic spatial patterns (HilleRisLambers et al. 2001, Rietkerk et al. 2002). An analysis of Rietkerk et al. (2002) illustrates that on a landscape level, alternative stable states between a state with vegetation patches and unvegetated land may occur over a range of rainfall levels. This is because in the absence of vegetation there is no local concentration of water, which is essential to allow vegetation persistence under low precipitation levels. Hence, once lost, vegetation

patches do not easily re-appear. An important bottom-line is that while external environmental heterogeneity can weaken catastrophic shifts and reduce hysteresis, self-organized heterogeneity in distribution of biota may in fact promote such phenomena (Rietkerk et al. 2004).

ACKNOWLEDGMENTS

We thank G. D. Peterson for his valuable comments on the manuscript.

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